

JACKSON NWS, KY, USA

WMO: 724190

Lat: 37.5911N

Lon: 83.3139W

Elev: 414

StdP: 96.45

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03889

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-9.4	-18.7	0.8	-9.6	-15.9	1.0	-7.7	7.9	4.1	7.1	4.2	1.9	240	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	31.9	23.0	30.7	22.7	29.5	22.2	24.8	29.7	24.0	28.5	23.4	27.7	1.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.9	27.7	22.6	18.2	26.8	22.1	17.6	25.9	77.2	29.7	74.3	28.5	71.6	27.8	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.3	4.6	-16.0	33.9	3.8	2.0	-18.8	35.4	-21.0	36.6	-23.1	37.7	-25.9	39.2		
(5)			-16.6	26.0	3.7	0.8	-19.2	26.6	-21.4	27.1	-23.4	27.5	-26.1	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.8	1.7	4.0	8.5	14.5	18.5	22.4	23.9	23.6	20.5	14.7	8.8	4.2
	DBStd	9.26	7.05	6.58	6.40	5.45	4.23	2.81	2.28	2.45	3.75	4.88	5.57	6.00
	HDD10.0	885	270	187	107	21	2	0	0	0	0	14	87	198
	HDD18.3	2301	514	402	308	137	53	5	1	1	22	133	288	438
	CDD10.0	2278	14	19	61	155	264	371	430	423	314	159	52	17
	CDD18.3	653	0	1	4	22	57	126	172	166	86	19	2	0
	CDH23.3	4502	0	2	26	165	376	846	1212	1151	599	118	8	0
	CDH26.7	1274	0	0	2	27	76	231	373	365	182	19	0	0
(14) Wind	WSAvg	1.8	2.4	2.4	2.5	2.4	1.6	1.2	1.0	1.0	1.1	1.5	2.0	2.2
(15) Precipitation	PrecAvg	1203	75	101	101	88	131	124	122	100	84	78	105	112
	PrecMax	1608	166	200	248	254	274	232	247	196	199	187	237	329
	PrecMin	873	20	32	37	20	57	35	56	39	28	13	37	44
	PrecStd	203	34	45	57	51	48	56	49	43	50	46	51	68
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	22.0	26.2	29.2	30.4	32.1	33.2	33.9	32.9	28.8	24.4	20.2
		MCWB	14.2	14.7	15.9	17.2	21.3	23.1	24.4	23.1	20.7	19.6	16.6	14.7
	2%	DB	17.1	19.0	23.4	27.2	28.7	30.6	31.4	31.7	30.7	26.3	21.6	17.4
		MCWB	12.8	12.7	14.3	16.3	20.0	22.8	23.9	23.2	21.1	18.4	14.3	13.0
	5%	DB	14.8	16.2	20.8	25.3	27.3	29.2	30.2	30.2	28.8	24.3	19.2	15.3
		MCWB	11.7	11.1	13.0	15.7	19.5	22.2	23.4	22.9	20.8	17.2	13.2	11.7
	10%	DB	12.3	13.7	18.2	23.2	25.6	27.9	28.8	28.7	26.9	22.3	17.2	13.3
		MCWB	9.0	9.5	11.7	14.6	18.9	21.5	22.8	22.3	19.9	16.3	12.1	10.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	16.2	16.8	19.5	22.9	25.0	25.8	25.4	23.6	22.0	17.8	15.9
		MCDB	18.2	20.3	23.7	25.9	28.2	30.0	30.9	30.9	29.1	26.5	22.7	18.8
	2%	WB	13.8	14.2	15.6	18.1	21.7	23.9	25.0	24.6	22.7	20.2	16.1	14.1
		MCDB	16.6	17.5	21.4	24.3	26.6	28.7	29.9	29.6	28.0	23.7	20.0	16.4
	5%	WB	12.0	12.2	14.3	16.9	20.7	23.1	24.3	23.8	21.7	18.8	14.5	12.5
		MCDB	14.5	14.9	19.2	22.5	25.4	27.8	28.5	28.4	26.4	22.8	18.0	14.7
	10%	WB	9.6	10.2	12.7	15.8	19.8	22.3	23.6	23.0	21.0	17.4	12.7	10.6
		MCDB	12.1	13.3	17.0	20.9	24.2	26.6	27.5	27.2	24.8	20.9	16.0	13.1
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	10.3	11.0	10.0	9.6	9.1	9.5	10.0	10.2	9.3	8.0
		MCDBR	10.1	12.3	12.7	12.9	11.4	10.9	10.6	11.0	12.0	12.2	11.1	10.3
		MCWBR	6.9	8.4	7.0	6.4	5.4	5.1	4.6	4.5	4.7	5.8	6.4	7.6
	5% WB	MCDBR	9.3	11.1	11.0	10.8	9.8	9.7	9.3	9.4	9.9	10.0	9.2	9.4
		MCWBR	7.1	8.8	7.0	6.1	5.2	5.0	4.6	4.4	4.5	5.5	6.6	7.7
(40) Clear-Sky Solar Irradiance	taub		0.300	0.312	0.337	0.382	0.427	0.444	0.471	0.478	0.396	0.351	0.321	0.297
	taud		2.520	2.478	2.450	2.326	2.208	2.201	2.148	2.115	2.329	2.459	2.513	2.542
	Ebn at Noon		893	923	927	898	858	841	814	798	852	865	859	872
	Edh at Noon		78	92	104	125	143	144	151	153	116	93	78	72
(44) All-Sky Solar Radiation	RadAvg		1.95	2.59	3.71	4.90	5.50	6.10	5.69	5.38	4.49	3.39	2.41	1.71
	RadStd		0.21	0.37	0.32	0.37	0.50	0.34	0.32	0.30	0.46	0.43	0.28	0.18

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+66
(47)	Regional (22 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+71

Nomenclature: See separate page